



CCRIMS

Cyber Crime Reporting & Investigation Management System

Database Systems

Group Members

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What is CCRIMS?

CCRIMS is a web-based platform designed to digitize and streamline the full lifecycle of cyber crime reporting and investigation — from citizen complaints to forensic analysis and legal compliance.



Purpose

Provide citizens a single platform to report cyber crimes and track their case in real time



Scope

Covers 8 crime types, 4 user roles, 13 DB tables, and a complete forensic evidence chain



Technology

Built on a relational database with BCNF normalization ensuring zero redundancy and full integrity

13 DB Tables | 4 User Roles | 8 Crime Types | BCNF Normalized

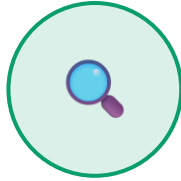
Roles, Modules & Objectives



Citizen

- ✓ Register & login securely
- ✓ File cyber crime complaints
- ✓ Track complaint status

13 DB Tables



Officer

- ✓ Manage & investigate cases
- ✓ Add investigation logs
- ✓ Manage suspects & warrants

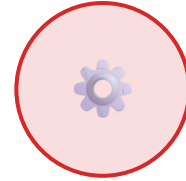
4 User Roles



Forensic Expert

- ✓ Collect digital evidence
- ✓ Submit forensic reports

8 Crime Types



Admin

- ✓ Manage all staff accounts
- ✓ View full system data
- ✓ Oversee all cases

3 Departments

ERD — CCRIMS Database



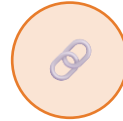
EERD — Key Enhancements Over Basic ERD



Specialization / Disjoint

Entity: STAFF

STAFF is specialized into 4 disjoint subclasses via the Role attribute. Each staff member belongs to exactly ONE role (disjoint, total participation).



Weak Entity

Entity: CASE_RECORD

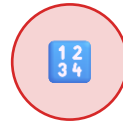
CASE_RECORD is a weak entity — it cannot exist without a parent CASE. Its identity depends on (CaseID + RecordDate + StaffID).



Aggregation

Entity: FORENSIC_REPORT

FORENSIC_REPORT is built on CASE_RECORD through aggregation — treating the CASE_RECORD relationship as a higher-level entity. RecordID is both PK and FK (1:1).



Multivalued Attr (removed)

Entity: Phone Numbers

Originally multivalued attributes for phone numbers were normalized — moved inline since each entity has at most one phone in this system.

First Normal Form (1NF)

RULE: Every column must be atomic · No repeating groups or arrays · Every row must have a Primary Key

✗ Repeating Phone Numbers

BEFORE:

```
CITIZEN(CitizenID, Fname, Lname, Email,  
Phone1, Phone2, Phone3)
```

AFTER:

```
CITIZEN(CitizenID, Fname, Lname, Email,  
Phone)  
→ Phone is now a single atomic column
```

✓ Merged Phone directly into CITIZEN & SUSPECT tables as single atomic column

✗ Non-Atomic Address

BEFORE:

```
SUSPECT(..., '123 Main St, Karachi,  
75000, Sindh, Pakistan')
```

AFTER:

```
SUSPECT(..., Street, City, ZipCode,  
State, Country)  
→ Each address component is atomic
```

✓ Address split into separate atomic columns — City stored directly in CITIZEN/SUSPECT

✗ Multi-Valued Evidence

BEFORE:

```
CASE(..., Evidence='log.txt, disk.img,  
screenshot.png')
```

AFTER:

```
DIGITAL_EVIDENCE(EvidenceID, FileName,  
EvidenceType, ...)  
→ Separate table per item
```

✓ Digital evidence extracted into its own table — one row per evidence item

Second Normal Form (2NF)

RULE: Must be in 1NF · No partial dependencies — every non-key attribute must depend on the FULL composite key

TABLE: CASE_STAFF (CaseID, StaffID, AssignedDate, AssignmentRole)

⚠ Does Role depend on only StaffID? → YES if Role = staff's own role

Renamed to AssignmentRole — confirms it means the role played ON THIS SPECIFIC CASE, so it depends on the full (CaseID, StaffID) key. ✅ No violation.

TABLE: CASE_SUSPECT (CaseID, SuspectID, InvolvementType)

⚠ Does InvolvementType depend on only CaseID or only SuspectID?

InvolvementType = how a suspect is involved IN THIS CASE → depends on full composite key (CaseID + SuspectID). ✅ No violation.

TABLE: COMPLAINT (ComplaintID, Description, LodgeDate, CitizenID, CrimeTypeID)

⚠ Could CrimeTypeName appear here instead of just CrimeTypeID?

CrimeTypeName would depend only on CrimeTypeID, not on ComplaintID. ❌
Partial dependency removed by keeping CYBER_CRIME_TYPE as a separate table.
✅ Fixed.

TABLE: CITIZEN_PHONE / SUSPECT_PHONE (were junction tables)

⚠ Original design had separate phone tables with composite PKs

Since each citizen/suspect has at most one phone in this system, Phone was merged directly into CITIZEN and SUSPECT — eliminating unnecessary junction tables. ✅ Simplified.

3NF & BCNF

Third Normal Form (3NF)

RULE: No transitive dependencies — non-key attributes must not depend on other non-key attributes.

CITIZEN / SUSPECT — Address

City is stored directly as atomic column. No $\text{ZipCode} \rightarrow \text{City}$ transitive dependency because ZipCode tables were removed from schema.

STAFF (StaffID $\rightarrow$ BadgeNumber)

Both StaffID and BadgeNumber are candidate keys. BadgeNumber declared UNIQUE. No transitivity — both are superkeys.

CASE — ComplaintID (1:1)

Complaint results in exactly one CASE. ComplaintID declared UNIQUE in CASE table making it a candidate key. ✓

FORENSIC_REPORT

RecordID is both PK and FK (1:1 with CASE_RECORD). No additional attributes create transitivity.

Boyce-Codd Normal Form (BCNF)

RULE: For every functional dependency $X \rightarrow Y$, X must be a superkey. Stricter than 3NF.

STAFF — Multiple Candidate Keys

StaffID, BadgeNumber, and Email are all candidate keys (UNIQUE). All are superkeys $\rightarrow$ no BCNF violation. ✓

MALWARE table — REMOVED

Hash $\rightarrow$ MalwareName was a dependency. Simplified by removing MALWARE as separate table — type stored in DIGITAL_EVIDENCE. ✓

DIGITAL_EVIDENCE — FileHash

FileHash declared UNIQUE making it a candidate key. FileHash $\rightarrow$ FileName is valid since FileHash is a superkey. ✓

FORENSIC_REPORT — 1:1 key

RecordID as PK ensures every FD has a superkey as determinant. No BCNF violation in any direction. ✓

What CCRIMS Achieves

Centralized Database

All cyber crime data — complaints, cases, suspects, evidence, reports — in one structured relational system

Role-Based Access

Citizen, Officer, Analyst and Admin each have dedicated secure interfaces with appropriate permissions

BCNF Normalized

13 tables designed through 1NF → 2NF → 3NF → BCNF. Zero redundancy, full referential integrity

Full Investigation Trail

Every case has a complete audit trail: case records, investigation logs, forensic reports and warrants

Digital Forensics

Digital evidence tracked with SHA-256 hash verification and chain of custody documentation

Legal Compliance

Warrant management with issue dates, expiry and status tracking ensures legal procedural compliance